NMC 2012 O-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn.1 What is half of 2^{2012}?

Qn.2 The function $f(x)$ satisfies the equation $(x-1)f(x) + f(\frac{1}{x}) = 1$ for all $x \neq 0$. Find the value of $f(2)$.

Qn.3 Two fractions are removed from the following six fractions $\frac{1}{3}, \frac{1}{6}, \frac{1}{9}, \frac{1}{12}, \frac{1}{15}, \frac{1}{18}$ so that the sum of the remaining four fractions is $\frac{2}{3}$. What is the product of the two fractions that were removed?

Qn.4 John takes three steps to walk the same distance as Jim walks in four steps. Each of John's steps covers 0.5 metres. How many metres does Jim travel in 24 steps?

Qn.5 In the diagram below, the numbers 1 to 10 are placed around a circle. Amina crosses out 1, then 4 and then 7. Counting in a clockwise direction, she crosses out every third number of those remaining, until only two numbers are left. What is the sum of these two numbers?

Qn.6 Find all the real numbers x for which the property that the median of five numbers x, 6, 4, 1, 9 is equal to their mean.

Qn.7 John and Joseph select T-shirts from a boutique. John only selects the ones with two colours and Joseph only selects the ones with seven colours. John has 5 T-shirts more than Joseph. The total number of colours found on all the selected T-shirts is 100. How many T-shirts did they select all together?

Qn.8 Without using tables or a calculator, find the last two digits of

$123456789 \times 98765342 \times 104392831$

Qn.9 Johann tells Kitty that the product of three positive integers is 36. Jonathan also tells Kitty the sum of the integers. However, on solving, Kitty found that the set of integers is not unique. What is the sum of these integers?

Qn.10 In the diagram below, PX is a tangent of a circle at A. PY is a tangent of the circle at B and CD is a tangent of the circle at E. If $\overline{PA} = 15$ units, find the perimeter of triangle PCD.

SECTION B

Qn.11 Two men Liam and Award went for a drink. Award took a cup filled with water and drank $\frac{1}{5}$ of its contents. When he looked aside, Liam re-filled the cup by adding alcohol and then stirred. Award drank $\frac{1}{4}$ of the mixture. When Award again looked aside, Liam re-filled the cup with more alcohol and stirred. Award drank $\frac{1}{3}$ of the mixture. When Award looked aside for the third time, Laim re-filled the cup with more alcohol. What percentage of the final mixture was alcohol?

Qn.12 Each of the 9 squares shown in the figure below is to contain one number chosen from numbers 1 to 9. No number is to be repeated. Suppose the sum of the numbers in the 5 squares aligned vertically is 32 and the sum of the numbers in the horizontal squares is 20. What number goes in the shared corner square?

Qn.13 David and Daniel are running at constant speeds in opposite directions around a circular track. David completes one lap every 56 seconds and meets Daniel every 24 seconds. How long does it take Daniel to complete one lap?

Qn.14 A point (x, y) is called an integral point if both x and y are integers. Find all the points on the graph of $\frac{1}{x} + \frac{1}{y} = \frac{1}{4}$ that are integral points. For example $(2, -4)$ is one such point.

Qn.15 (a) Expand $(x - a)(x - b)(x - c) \ldots (x - z)$.

(b) Modify the right angled triangle with sides 11, 60 and 61 by doubling the length of its shortest side but keeping its perimeter and largest angle unchanged. What is the length of the new triangle's hypotenuse?

NMC 2012 O-LEVEL PAPER 2 SOLUTIONS

SECTION A

1. $\frac{1}{2} \times 2^{2012} = \frac{1}{2} \times 2 \times 2^{2011} = 2^{2011}$.

2. Substituting $x = 2$ we get $(2-1)f(2) + f\left(\frac{1}{2}\right) = 1$ or

$$f(2) + f\left(\frac{1}{2}\right) = 1. \tag{7}$$

Likewise, substituting $x = \frac{1}{2}$ gives $(\frac{1}{2} - 1)f\left(\frac{1}{2}\right) + f(2) = 1$ or

$$-\frac{1}{2} \cdot f\left(\frac{1}{2}\right) + f(2) = 1. \tag{8}$$

Solving equations (7) and (8) simultaneously (i.e $(\frac{1}{2})\times$ (7)+ (8)) gives $\frac{3}{2}f(2)+\frac{3}{2}$ giving $f(2) = 1$.

3.

$$\frac{1}{3}+\frac{1}{6}+\frac{1}{9}+\frac{1}{12}+\frac{1}{15}+\frac{1}{18} = \frac{60+30+20+15+12+10}{180} = \frac{147}{180}$$

$$\frac{147}{180} = \frac{120+27}{180} = \frac{120}{180}+\frac{27}{180} = \frac{2}{3}+\frac{27}{180}.$$

The fraction that give a sum of $\frac{27}{180}$ are $\frac{1}{12}$ and $\frac{1}{15}$.
Therefore, the fractions removed are $\frac{1}{12}$ and $\frac{1}{15}$. Their product is $\frac{1}{12} \times \frac{1}{15} = \frac{1}{180}$.

4. The distance covered by Jim in 4 steps is the same distance covered by John in 3 steps. This implies that the distance covered by Jim in 24 steps (4×6) is the same distance covered by john in $(3 \times 6) = 18$steps. Since each of John's steps is $0.5m$,, then Jim covers $18 \times 0.5 = 9$ metres in 24 steps.

5. The sum of the two numbers is $8 + 2 = 10$.

6. • Suppose that x is the less than 4. This implies that the median is 4. Then,

$$4 = \frac{x+6+4+1+9}{5} \Rightarrow 20 = x + 20 \Rightarrow x = 0.$$

- Suppose that x is between 4 and 6. This implies that the median is x. Then, then

$$x = \frac{x+6+4+1+9}{5} \Rightarrow 5x = x+20 \Rightarrow 4x = 20 \Rightarrow x = 5.$$

- Suppose that x is greater than 6. This implies that the median is 6. Then,

$$6 = \frac{x+6+4+1+9}{5} \Rightarrow 30 = x+20 \Rightarrow x = 10.$$

The only possible values of x are $0, 5$ and 10.

7. Let x be the number of T-shirts that John selects and let y be the number of T-shirts that Joseph selects. We are given that

$$2x + 7y = 100 \quad (9)$$
$$x - y = 5 \quad (10)$$

(9) minus $2\times$ (10) gives $y = 10$ and $x = 15$.

Therefore, they all together select $x + y = 25$ shirts.

8. $123456789 \times 98765342 \times 104392831$.

The last two digits of any product is given by the product of the any two last digits of each number. i.e. $89 \times 42 \times 31 = 3738 \times 31 = 115878$. Therefore, the last two digits are 78.

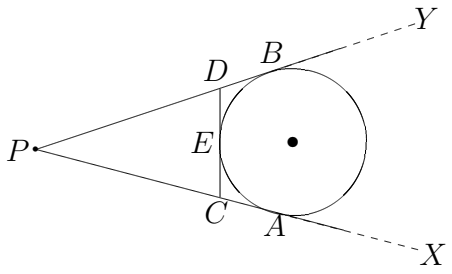
9. The set of all possible 3 positive integers whose product is 36 is

$$\{1,1,36\},\{1,2,18\},\{1,3,12\},\{1,4,9\},\{1,6,6\},\{2,2,9\}\{2,3,6\},\{3,3,4\}.$$

The sum of the elements in all sets are different except for $\{1,6,66\}$ and $\{2,2,9\}$. Since Kitty found out that the sum of integers was not unique, then, it must be one of these sets whose sum is 13.

NB: In enumerating the sets, we have excluded sets of the same elements which are arranged differently. For example, sets such $\{1,4,9\},\{4,9,1\},\{4,1,9\},\{9,4,1\}$ and $\{9,1,4\}$ have the same elements and can be taken to be equal and same sets.

10.



$\overline{PA} = \overline{PB} = 15$ (Tangent from a common point P)

Similarly $\overline{CA} = \overline{CE}$(tangent from a common point C and $\overline{DB} = \overline{DE}$ (Tangent from a common point D).

$$\begin{aligned}
\text{Perimeter of } \triangle PCD &= \overline{PC} + \overline{CE} + \overline{ED} + \overline{DP} \ (\overline{CD} = \overline{CE} + \overline{ED}). \\
&= \overline{PC} + \overline{CA} + \overline{DB} + \overline{DP} \\
&= \overline{PA} + \overline{PB} \\
&= 15 + 15 = 30cm.
\end{aligned}$$

SECTION B

11. Let A be the amount of alcohol present in the mixture at sometime. Let p be the fraction of the mixture that Award drinks.

    This implies that the amount of alcohol taken by Award is Ap and the amount of alcohol left in the cup is $A - Ap = A(1-p)$..

    At the start, Award drinks $\frac{1}{5}$ of the water. Liam adds $\frac{1}{5}$ of the alcohol to the cup. When Award drinks $\frac{1}{4}$ of the mixture, it implies that the amount of alcohol left is $\frac{1}{5}\left(1 - 1\frac{1}{4}\right) = \frac{1}{5} \times \frac{3}{4} = \frac{3}{20}$.

    When Liam replaces with $\frac{1}{4}$ of alcohol, it implies that the amount of alcohol in the cup is $\frac{3}{20} + \frac{1}{4} = \frac{3+5}{20} = \frac{8}{20}$.

    When Award drinks $\frac{1}{3}$ of the mixture, then, the amount of alcohol left is $\frac{8}{20}\left(1 - \frac{1}{3}\right) = \frac{8}{20} \times \frac{2}{3} = \frac{16}{60}$.

    On the third time Liam replaces with $\frac{1}{3}$ of the alcohol to fill the cup. This implies that the amount of alcohol in the cup is $\frac{16}{60} + \frac{1}{3} = \frac{16+20}{60} = \frac{36}{60}$.

    Therefore the percentage of alcohol in the final mixture is $\frac{36}{60} \times 100\% = 60\%$.

12. Let x be the number in the shared corner square. The sum of all the numbers in the squares is equal to the sum of the numbers in the five squares aligned vertically plus the sum of the numbers in the horizontal squares minus x(because it is repeated). i.e.

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 32 + 20 - x \;\Rightarrow\; x = 7.$$

    Therefore, the number that goes in the shared corner square is 7.

13. It takes David $56s$ to complete one lap. In $1s$ he completes $\frac{1}{56}$ of the lap. Therefore in $24s$, David completes $\frac{24}{56} = \frac{3}{7}$ of the track.

    Daniel is running in the opposite direction. He will have to move $(1 - \frac{3}{7}) = \frac{4}{7}$ of the track in $24s$.

    Therefore, to complete one track $(\frac{7}{7})$, Daniel will have to take $\frac{24 \times 7}{4} = 42s$.

14.

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{4} \;\Rightarrow\; \frac{1}{y} = \frac{1}{4} - \frac{1}{x} = \frac{x-4}{4x}.$$

    Therefore, $y = \frac{4x}{x-4}$, $x \neq 0$ and $y \neq 0$.

    - when $x = \pm 1$, $y = \frac{4}{-3}$ (or $\frac{-4}{-1-4} = \frac{4}{5}$) which is not an integer. Likewise, when $x = \pm 2, y$ turns out not to be an integer. In the same way, when $x = -3$, the $y-$coordinate turns to be be non-integer. The same applies for $x = 4, x = -5, x = -6, x = \pm 7, x = -8, x = \pm 9, x = \pm 10$.

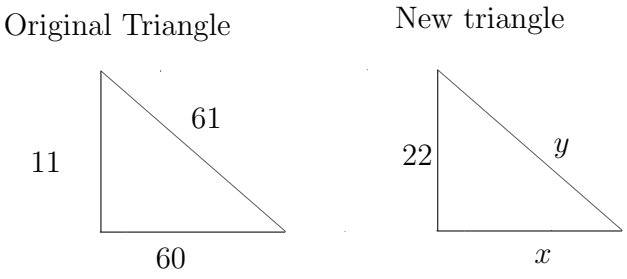
- When $x = 3, y = \frac{4\times3}{3-4} = -12$ implying that $(3, -12)$ is a solution.
- When $x = -4, y = \frac{4\times-4}{-4-4} = 2$ implying that $(-4, 2)$ is a solution.
- When $x = 6, y = \frac{4\times6}{6-4} = 12$ implying that $(6, 12)$ is a solution.
- When $x = 8, y = \frac{4\times8}{8-4} = 8$ implying that $(8, 8)$ is a solution.

Expected Solutions are $\{(-12, 3), (-4, 2), (2, -4), (3, -12), (5, 20), (6, 12), (8, 8), (12, 60)$ and $(20, 5)$.

15. (a)

$$(x-a)(x-b)(x-c)\ldots(x-z) = (x-a)(x-b)(x-c)(x-d)(x-e)\ldots(x-x)(x-y)(x-z) = 0.$$

(b)



The original triangle has perimeter$11 + 60 + 61 = 12$ units.

Let the other two sides of the new triangle be x and y respectively. This implies that $p = x + y + 22 = 132$ since the perimeter remains the same. Making y the same subject gives $y = 110 - x$.

Using the new triangle, we have

$$\begin{aligned} x^2 + 22^2 &= (110 - x)^2 \\ \Rightarrow\ x^2 + 484 &= 12100 - 220 + x^2 \\ \Rightarrow\ -11616 &= -220x \\ \Rightarrow\ x &= 52.8. \end{aligned}$$

Thus, the length of the hypotenuse of the new triangle is $110 - x = 110 - 52.8 = 57.2$ units.